University of Baghdad
College of Medicine2025-2026

Title: Introductions to Respiratory Medicine

Grade: 4th grade

Module: Respiratory Medicine Module

Speaker: Prof. Dr: Mohammed Waheeb

Date:

Objectives

At the end of this lecture, the student should be able to:

- Recognize the different types of investigations used in respiratory diseases.
- Relate between common respiratory diseases and their specific investigation.
- Identify investigational abnormality in some common respiratory disorders.

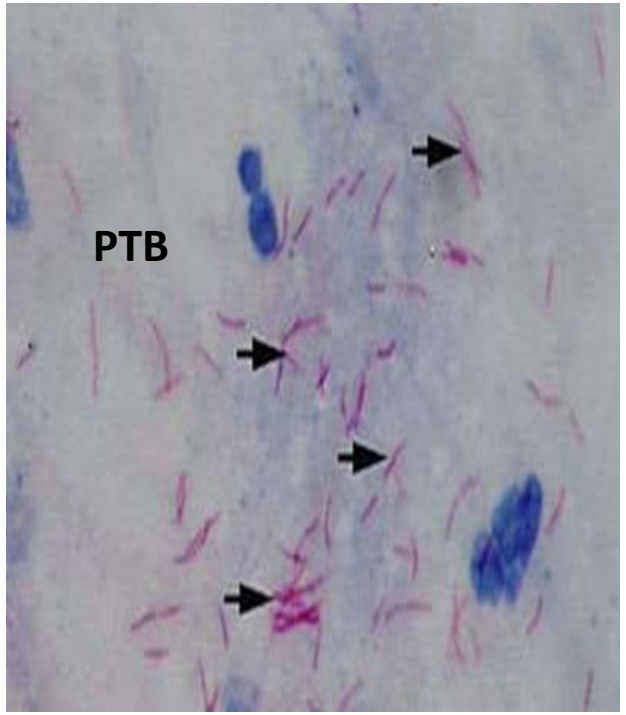
Investigation of respiratory disease include:

- Sputum Analysis
- Radiology
- Pulmonary function testing
- Oximetry and Arterial blood gas analysis
- Endoscopic examination
- Investigation for pleural diseases

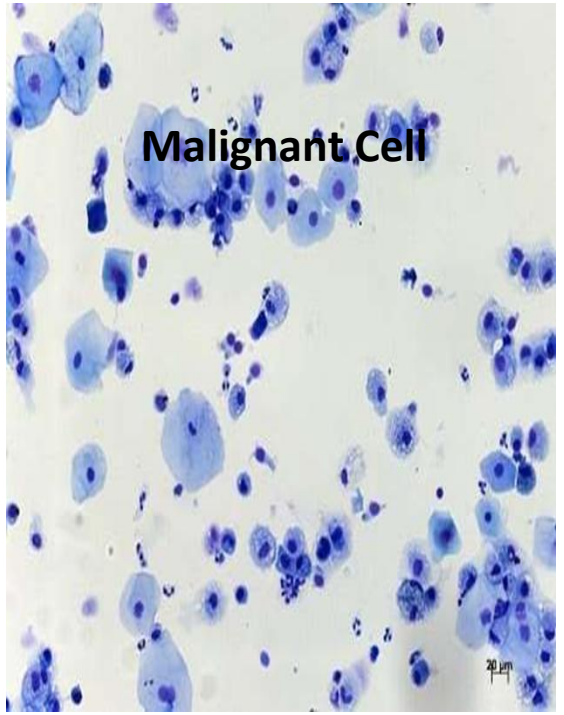
Sputum sample collection

- **For bacterial and fungal infection. e.g. for tuberculosis (commonly tested in Iraq).**
- **Cytology test used when bronchial cancer suspected**

PTB







Radiology

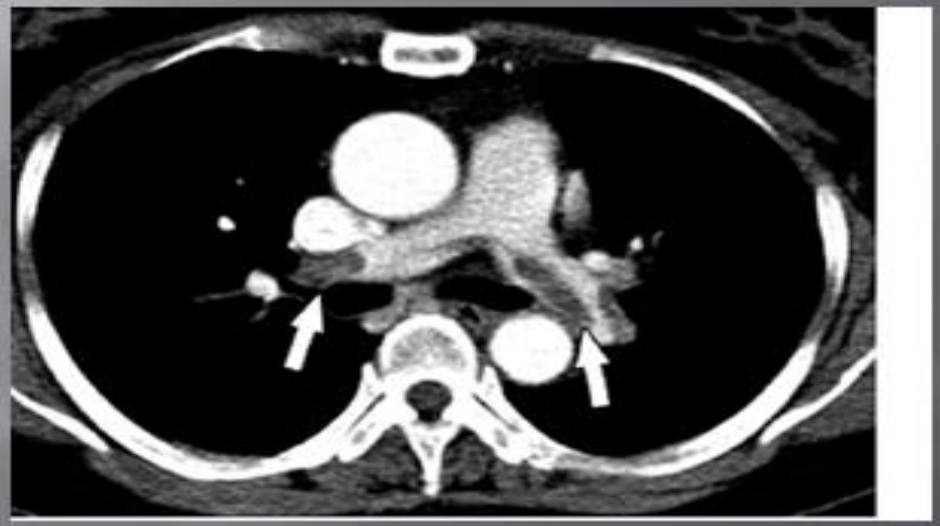
1. Chest X-ray
2. Computed tomography (CT)
3. Ultrasound
4. Ventilation–perfusion imaging
5. Positron emission tomography (PET)

1. Chest X-ray

This is performed on the majority of patients suspected of having chest disease provides information on the lung fields, heart, mediastinum, vascular structures and the thoracic cage

2- Computed Tomography (CT)

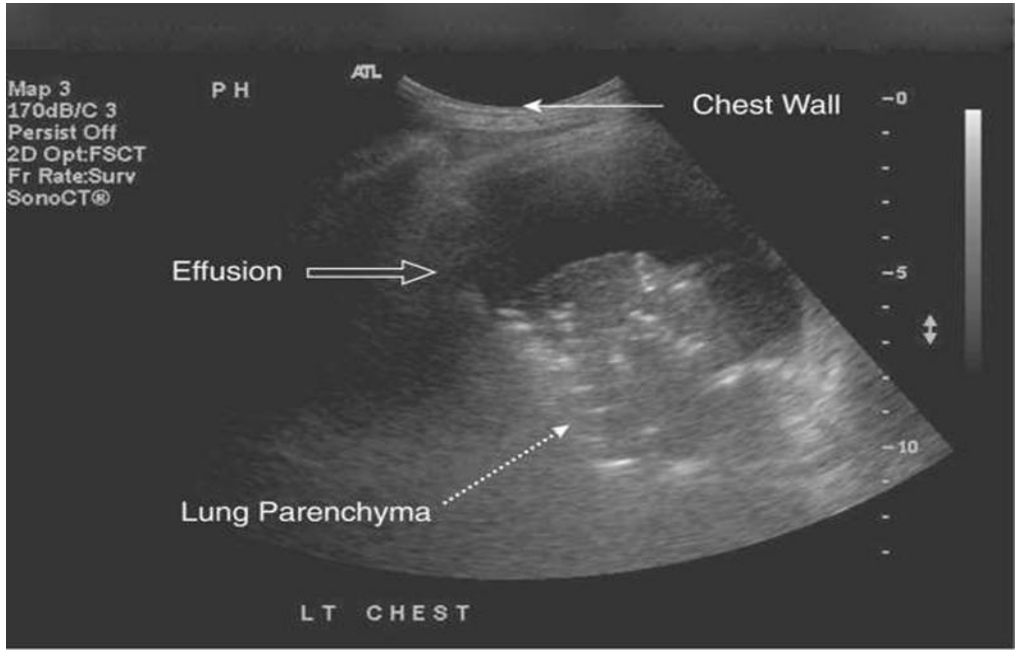
- **CT provides detailed images of the pulmonary parenchyma, mediastinum, pleura and bony structures.**
- **Normal and abnormal structures will be seen "clearly" by CT.**
- **Sampling of lung/pleural mass could be guided by CT.**



Pulmonary artery thrombosis

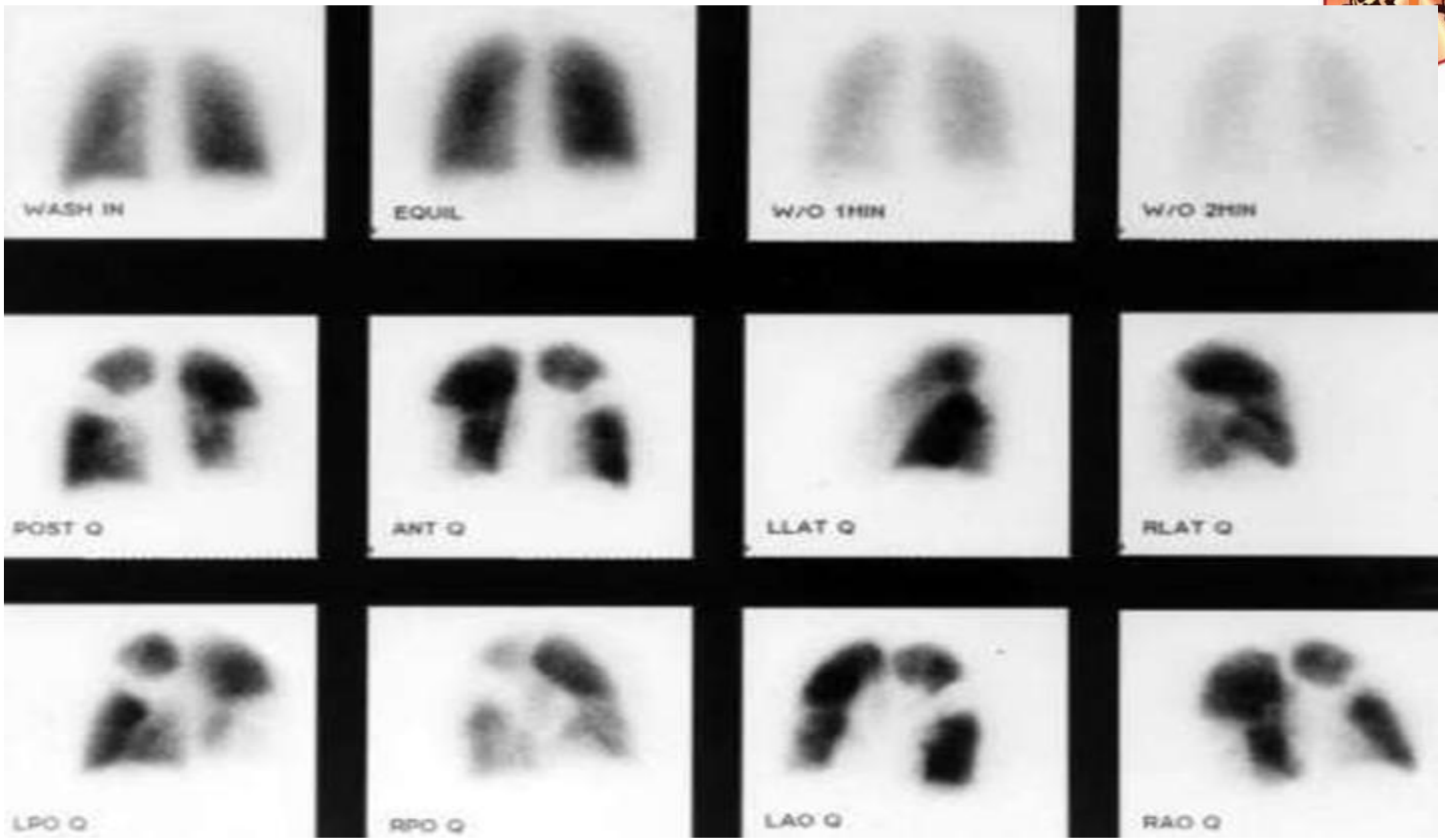
3-Ultrasound

- Ultrasound (US) is sensitive at detecting pleural fluid (effusion)



4- Ventilation–perfusion imaging

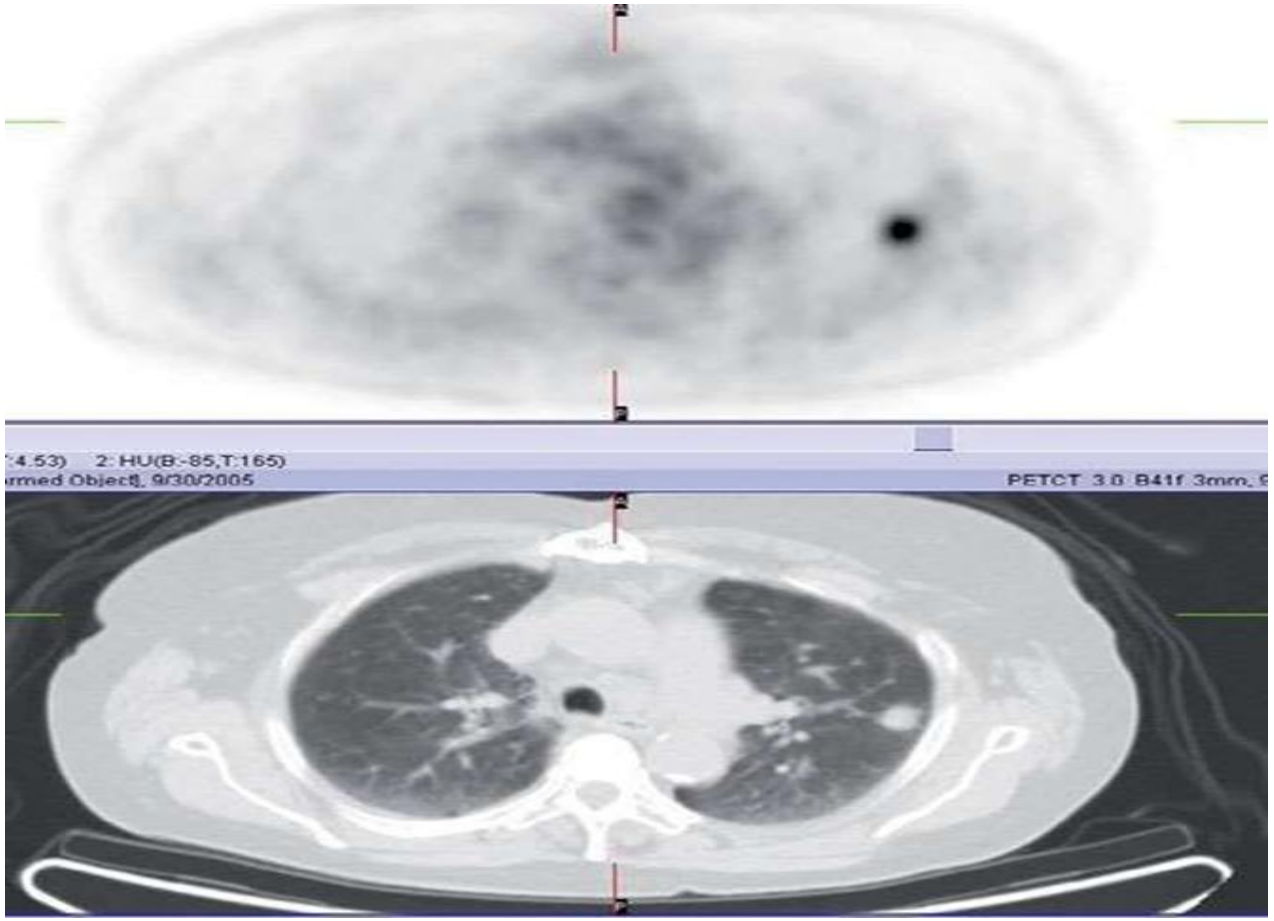
- **In this technique, the lungs are imaged using a gamma camera.**
- **Pulmonary embolism could be detected with this test.**
- **This test has been replaced by CT chest**


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5-Positron emission tomography (PET)

- **PET scanners show the avid ability of malignant tissue to absorb and metabolize glucose.**

- **Used largely in detecting malignant tumor metastasis.**


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PETCT 3.0 B41f 3mm

Endoscopic examination

1. **Bronchoscopy**
2. **Mediastinoscopy**
3. **Laryngoscopy**
4. **Thoracoscopy.**
5. **Pleuroscope.**

Pulmonary function testing

- Respiratory function tests are used to aid diagnosis, assess functional impairment, and monitor treatment or progression of disease.
- Obstructive (e.g. asthma) and restrictive lung diseases (e.g. pulmonary fibrosis)could be assessed with these tests.

Oximetry and Arterial blood gas analysis

- Oximeter is a an instrument that can measure O2 saturation.
- Arterial blood gas machine measure O2 andCO2pH and bicarbonate.
- Respiratory system function is (gas exchange), this test give good idea about this function.
- The test will show:

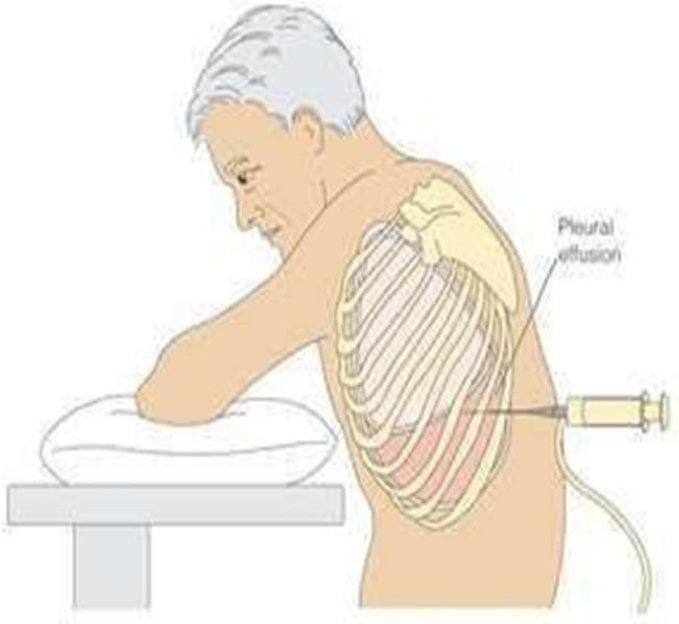
O2, CO2,pH& bicarbonate level.

Investigation for pleural diseases

- Pleural fluid aspiration
- Pleural tissue biopsy
- Pleuroscope.

Pleural fluid aspiration

- Aspiration from abnormal pleural fluid collection (pleural effusion) is achieved by needle and this sample analyzed for color, protein, bacteriology, cytology and many other tests accordingly.



Pleural tissue biopsy

- Biopsy of the pleura is achieved with Abram needle or Thoracoscopy and this tissue sample checked for histological abnormality e.g. tuberculosis and malignancy



Abram needle

Summary

- **Respiratory diseases are common.**
- **Investigations used for this important system include sputum sampling, radiology, pulmonary function tests, oximetry, blood gas analysis and endoscopy.**
- **It is important to identify investigational abnormality in of the common diseases**

Reference

- **Davidsons Principles and Practice of Medicine**



Thank you